KAIZEN MATHS

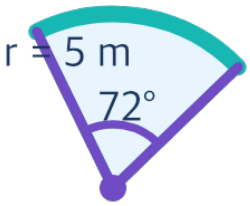
Lesson starter

Sectors, Arc Length and Area

Arc Length

Name _____

SA1



Find L

GEO-2EB5-L1 · SA

Show your working

KAIZEN MATHS

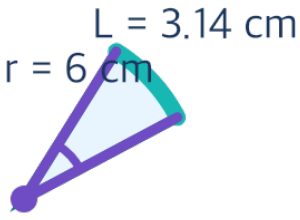
Lesson starter

Sectors, Arc Length and Area

Arc Length

Name _____

SB1



Find θ

GEO-2EB5-L1 · SB

Show your working

KAIZEN MATHS

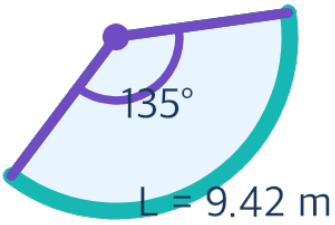
Lesson starter

Sectors, Arc Length and Area

Arc Length

Name _____

SC1



Find r

GEO-2EB5-L1 · SC

Show your working

KAIZEN MATHS

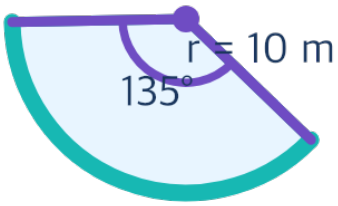
Lesson starter

Sectors, Arc Length and Area

Arc Length

Name _____

SD1



Find L

GEO-2EB5-L1 · SD

Show your working

KAIZEN MATHS

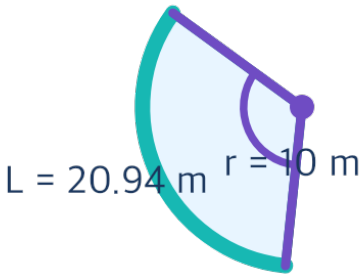
Exit ticket

Sectors, Arc Length and Area

Arc Length

Name _____

EA1



Find θ

GEO-2EB5-L1 · EA

Show your working

KAIZEN MATHS

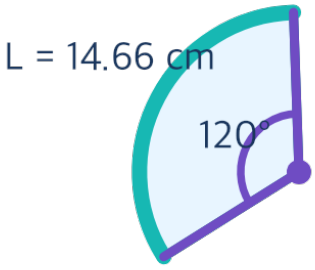
Exit ticket

Sectors, Arc Length and Area

Arc Length

Name _____

EB1



Find r

GEO-2EB5-L1 · EB

Show your working

KAIZEN MATHS

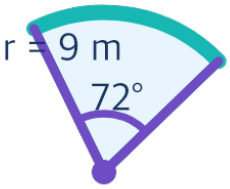
Exit ticket

Sectors, Arc Length and Area

Arc Length

Name _____

EC1



Find L

GEO-2EB5-L1 · EC

Show your working

KAIZEN MATHS

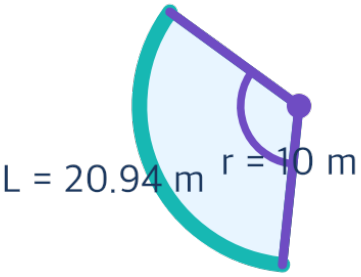
Exit ticket

Sectors, Arc Length and Area

Arc Length

Name _____

EA1



Find θ

GEO-2EB5-L1 · EA

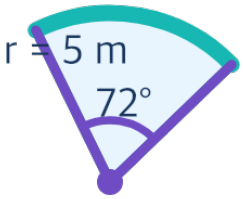
Show your working

Teacher answers & working

Sectors, Arc Length and Area · Arc Length

GEO-2EB5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Find L

Answer:

6.28 m

Working:

$L = \theta/360 \times 2\pi r$

$L = 72/360 \times 2\pi \times 5$

$L = 2\pi$ m

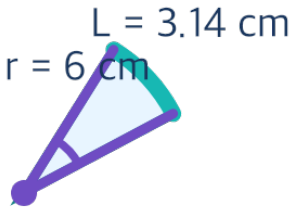
$L \approx 6.28$ m.

Teacher answers & working

Sectors, Arc Length and Area · Arc Length

GEO-2EB5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Find θ

Answer:

30°

Working:

$L = \theta/360 \times 2\pi r$

$\theta = L \div (2\pi r) \times 360$

$\theta \approx 3.14 \div (2\pi \times 6) \times 360$

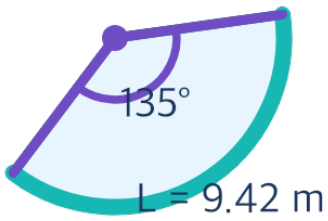
$\theta \approx 30°$.

Teacher answers & working

Sectors, Arc Length and Area · Arc Length

GEO-2EB5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Find r

Answer:

4 m

Working:

$L = \theta/360 \times 2\pi r$

$r = L \div (\theta/360 \times 2\pi)$

$r \approx 9.42 \div (135/360 \times 2\pi)$

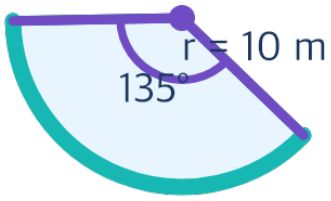
$r \approx 4$ m.

Teacher answers & working

Sectors, Arc Length and Area · Arc Length

GEO-2EB5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Find L

Answer:

23.56 m

Working:

$L = \theta/360 \times 2\pi r$

$L = 135/360 \times 2\pi \times 10$

$L = 15\pi/2$ m

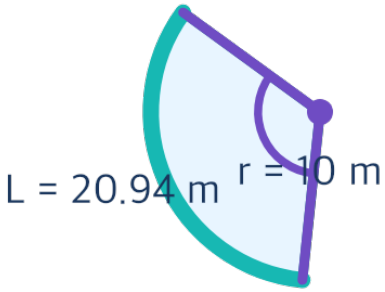
$L \approx 23.56$ m.

Teacher answers & working

Sectors, Arc Length and Area · Arc Length

GEO-2EB5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Find θ

Answer:

120°

Working:

$L = \theta/360 \times 2\pi r$

$\theta = L \div (2\pi r) \times 360$

$\theta \approx 20.94 \div (2\pi \times 10) \times 360$

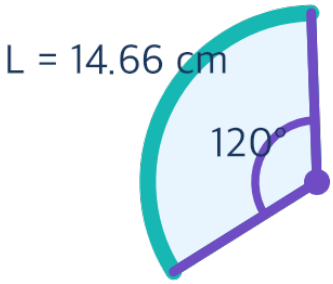
$\theta \approx 120°$.

Teacher answers & working

Sectors, Arc Length and Area · Arc Length

GEO-2EB5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Find r

Answer:

7 cm

Working:

$L = \theta/360 \times 2\pi r$

$r = L \div (\theta/360 \times 2\pi)$

$r \approx 14.66 \div (120/360 \times 2\pi)$

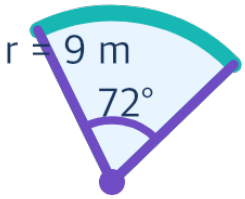
$r \approx 7$ cm.

Teacher answers & working

Sectors, Arc Length and Area · Arc Length

GEO-2EB5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Find L

Answer:

11.31 m

Working:

$L = \theta/360 \times 2\pi r$

$L = 72/360 \times 2\pi \times 9$

$L = 18\pi/5$ m

$L \approx 11.31$ m.